BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI
Hyderabad Campus
SECOND SEMESTER 2025-2026

Mathematical Foundations for Data Science

Assignment - 2

Max Marks: 10 Last Date for submission: 16-April-2026

Instructions:

- Use Python coding to solve the following problems
- For each problem, the detailed python coding, the final and intermediate solutions page has to be attached
- Make a single PDF file and submit
- Write your name and ID number at the top of the first page
- First two questions carries 4M each and the last question carries 2M

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1. Gradient Descent with Adam optimizer for the problem having objective function $f(x, y) = x^2 + y^2$ with the following settings: $x_0 = -2.0, y_0 = 2.0$, learning rate $\alpha = 0.1$, $\rho = \rho_f = 0.9$, $\epsilon = 10^{-8}$, max epochs = 50.
2. Gradient Descent with AdaGrad optimizer for the problem having objective function $f(x, y) = x^2 + y^2$ with the following settings: $x_0 = -2.0, y_0 = 2.0$, learning rate $\alpha = 0.1$, $\epsilon = 10^{-8}$, max epochs = 50.
3. Using Lagrange multipliers to minimize the function $f(x, y) = x + 2y$ subject to $x^2 + y^2 = 4$. Take initial guess $(0, 0)$.